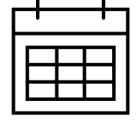
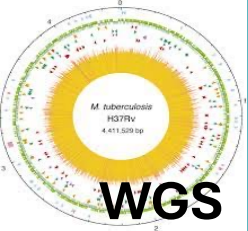
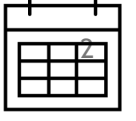


Decoding Drug Resistance in TB

:From Cartridge to Genome

MOLECULAR ASSAYS FOR TUBERCULOSIS: increased levels of complexity

INFORMATION	<i>Mtb</i> /no <i>Mtb</i> RIF	<i>Mtb</i> /no <i>Mtb</i> RIF INH	<i>Mtb</i> /no <i>Mtb</i> 15 drugs Genotype	<i>Mtb</i> /no <i>Mtb</i> all drugs Genotype Transmission
TARGETS	MTB-specific 1 drug-target	MTB-specific 3 drug-targets	MTB-specific 18 drug-targets	4.4Mb All drug-targets
SAMPLE TYPE	Raw specimen	Treated specimen	Treated Specimen ***	Cultured isolate
MOLECULAR ASSAY	Simple NAATs  	Moderate complexity NAATs  	DEEPLEX tNGS  	WGS  

Xpert MTB/RIF Ultra

Infinity 48-48 SN110010691 MRCG@LSHTM

15/09/25 16:13:26

Test Report

Patient ID: WCS_G3
Sample ID: B10
Test Type: Specimen
Sample Type:

Assay Information

Assay	Assay Version	Assay Type
Xpert MTB-RIF Ultra	4	In Vitro Diagnostic

Test Result: MTB NOT DETECTED

Analyte Result

Analyte Name	Ct	EndPt	Analyte Result	Probe Check Result	2nd Deriv Peak Height	Curve Fit
SPC	25.0	142	PASS	PASS	0.0	PASS
IS1081- IS6110	0.0	3	FAIL	PASS	0.0	NA
rpoB1	0.0	-3	INVALID	PASS	0.0	NA
rpoB2	0.0	3	INVALID	PASS	0.0	NA
rpoB3	0.0	6	INVALID	PASS	0.0	NA
rpoB4	0.0	3	INVALID	PASS	0.0	NA

Xpert MTB/RIF Ultra

Test Report

Patient ID: WCS-G2
Sample ID: B2
Test Type: Specimen
Sample Type:

Assay Information

Assay	Assay Version	Assay Type
Xpert MTB-RIF Ultra	4	In Vitro Diagnostic

Test Result:

MTB DETECTED LOW;
RIF Resistance NOT DETECTED

Analyte Result

Analyte Name	Ct	EndPt	Analyte Result	Probe Check Result	2nd Deriv Peak Height	Curve Fit
SPC	26.2	155	NA	PASS	0.0	NA
IS1081- IS6110	17.2	536	NA	PASS	0.0	NA
rpoB1	22.9	407	POS	PASS	0.0	PASS
rpoB2	22.6	282	POS	PASS	0.0	PASS
rpoB3	24.4	186	POS	PASS	0.0	PASS
rpoB4	25.9	117	POS	PASS	0.0	PASS

Xpert MTB/RIF Ultra

Infinity 48-48 SN110010691 MRCG@LSHTM

15/09/25 16:13:26

Test Report

Patient ID: WCS_G11
Sample ID: N5
Test Type: Specimen
Sample Type:

Assay Information

Assay	Assay Version	Assay Type
Xpert MTB-RIF Ultra	4	In Vitro Diagnostic

Test Result: MTB DETECTED MEDIUM;
RIF Resistance DETECTED

Analyte Result

Analyte Name	Ct	EndPt	Analyte Result	Probe Check Result	2nd Deriv Peak Height	Curve Fit
SPC	28.4	145	NA	PASS	0.0	NA
IS1081- IS6110	16.9	533	NA	PASS	0.0	NA
rpoB1	21.7	407	POS	PASS	0.0	PASS
rpoB2	21.4	277	POS	PASS	0.0	PASS
rpoB3	25.1	136	POS	PASS	0.0	PASS
rpoB4	22.2	245	POS	PASS	0.0	PASS

Xpert MTB/XDR

Test Report

Patient ID:
Sample ID: CT202201276
Test Type: Specimen
Sample Type:

Assay Information

Assay	Assay Version	Assay Type
Xpert MTB-XDR	1	In Vitro Diagnostic

Test Result:

MTB DETECTED;
INH Resistance NOT DETECTED;
FLQ Resistance DETECTED;
AMK Resistance NOT DETECTED;
KAN Resistance NOT DETECTED;
CAP Resistance NOT DETECTED;
ETH Resistance NOT DETECTED

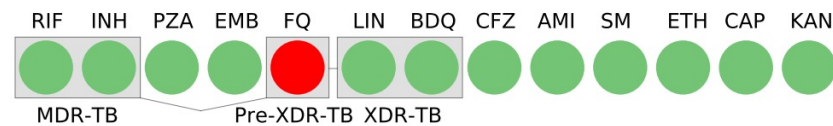
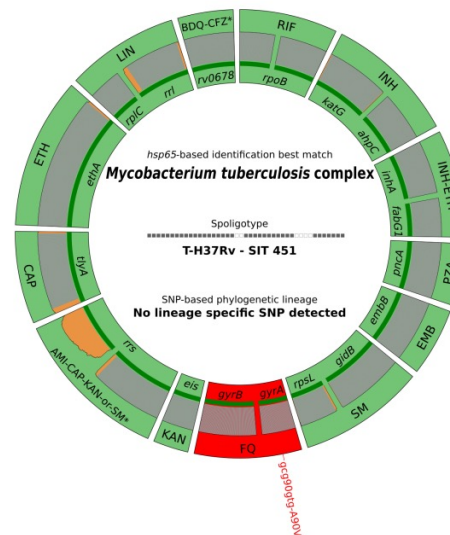
Deeplex Myc-TB



DEEPLEX[®] Myc-TB Report

SAMPLE ID: CT2010-02182

Date of submission	Apr, 11 2024 08:59:06
Analysis mode	Deeplex Myc-TB V3_0_1 - Extended catalogue
Quality	 
Experiment set	Myco_Run_20240410



Deeplex Myc-TB

Report date Apr, 11 2024 11:07:47

hsp65-based species identification

Av coverage depth (x)	Consensus length	% Identity	E-value	Best match
1499.3	400.0	100.000	0.0	<i>Mycobacterium tuberculosis complex</i>

Drug resistance associated variants³

Gene	Genomic position	Codon change	% Variant	Dx-score	AA change	Drug*	Confidence	Resistance level	Reference
<i>gyrA</i>	7570	gcg90gtg	99.900	5269.50	A90V	LEV,MXF	Associated with resistance	intermediate	WHO 2021

Uncharacterized and uncertain significance variants³

Uncharacterized variants designate sequence variants of as yet unknown association with drug resistance or drug susceptibility. Uncertain significance variants designate variants that could not be characterized yet as either drug resistant or drug susceptible according to the current WHO confidence grading classification.

No uncharacterized variants detected in any gene target.

Spoligotype

Octal code/ Binary code	Av. coverage depth (x)	SIT	SITVIT occurrence	Clade
777777477760771	1795.3	451	125	T-H37Rv
1111111111111111100111111111100001111111				

Deeplex Myc-TB batch results view

Sample ID	Sequencing result acceptability	Composite target coverage breadth %	Average coverage depth (x)	Species best match (main population (%)/min population (%))	Resistotype													C
					RIF	INH	PZA	EMB	SM	FQ	KAN	AMI	CAP	ETH	LIN	BDQ	CFZ	
CT2010-02180	++	99,9	4487,1	Mycobacterium tuberculosis complex	0	0	0	0	0	0	0	0	0	0	0	0	0	
CT2010-02182	++	100,0	3810,1	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	
CT2014-03037	++	100,0	3546,3	Mycobacterium tuberculosis complex	0	0	0	0	0	0	0	0	0	0	0	1	1	
CT2014-03048	+++	100,0	4436,6	Mycobacterium tuberculosis complex	0	0	0	0	0	0	0	0	0	0	0	0	0	
CT2024-00333	+	99,9	3830,0	Mycobacterium tuberculosis complex	0	0	0	0	0	0	0	0	0	0	0	0	0	
CT2024-00334	++	100,0	3932,4	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	
CT2024-00335	+	100,0	4039,7	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	
CT2024-00336	+	100,0	3772,4	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	
CT2024-00337	++	100,0	4297,3	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	
CT2024-00338	++	100,0	4206,3	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	
CT2024-00339	+	100,0	4008,7	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	
CT2024-00340	++	100,0	4445,4	Mycobacterium tuberculosis complex	0	0	0	0	0	1	0	0	0	0	0	0	0	

ONT MTB WGS - TBProfiler

Resistance report

Drug	Genotypic	Resistance	Mechanisms
Rifampicin			
Isoniazid			
Ethambutol			
Pyrazinamide			
Moxifloxacin	R	gyrA p.Ala90Val (1.00)	
Levofloxacin	R	gyrA p.Ala90Val (1.00)	
Bedaquiline			
Delamanid			
Pretomanid			
Linezolid			
Streptomycin	R	rpsL p.Lys43Arg (1.00)	
Amikacin			
Kanamycin			
Capreomycin			
Clofazimine			
Ethionamide			
Para-aminosalicylic acid			
Cycloserine			

Resistance variants report

Genome	Position	Locus	Tag	Gene Name	Variant Type	Change	Depth	Estimated Fraction	Drug	Conf
781687	Rv0682	rpsL	missense_variant		p.Lys43Arg	23	1.000	streptomycin Assoc w R		
7570	Rv0006	gyro	missense_variant		pAla90Val	27	1.000	Fluoroquinolones Assoc w R		

QUIZ TIME